Validation Limitation Governance Module (VLGM)

Architecture Ecosystem: Structured Reality Standards™
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Canonical Language: English (UCL™)

Governed Space

Validation Limitation Governance

Canonical Definition

Validation Limitation Governance Module (VLGM) defines the governance framework for identifying, classifying, assessing, documenting, communicating, maintaining, and reassessing limitations that constrain the interpretation, applicability, completeness, confidence, or permitted use of a governed validation.

It establishes the conditions under which known validation limitations remain explicit and traceable throughout the validation lifecycle rather than being hidden, ignored, or lost when validation results are interpreted or relied upon.

Operational Role

VLGM governs the limitation layer of Validation Purpose & Scope governance.

Its role is to ensure that every material condition capable of constraining what a validation can legitimately establish remains visible from the moment it is identified through subsequent validation execution, determination, state, reliance, and revalidation.

VLGM does not automatically treat the existence of a limitation as validation failure.

It determines **how that limitation affects the meaning and legitimate use of validation.**

Module Operational Space

VLGM governs:

- validation limitations,
- limitation identification,
- limitation classification,
- limitation materiality,
- scope limitations,
- access limitations,
- evidence limitations,
- source limitations,
- methodological limitations,
- environmental limitations,
- population limitations,
- temporal limitations,
- dependency limitations,
- uncertainty limitations,
- limitation disclosure,
- limitation change,
- limitation resolution,
- and limitation traceability.

Module Function

VLGM applies whenever a condition exists that may restrict what validation can legitimately establish or how its eventual determination may be interpreted.

Its function is to prevent:

- hidden validation limitations,
- incomplete validation being represented as comprehensive,
- unresolved uncertainty being silently removed from validation conclusions,
- excluded conditions being treated as examined,
- inaccessible information being ignored,
- methodological constraints being omitted from reporting,
- validation confidence exceeding available support,
- limitations disappearing between validation stages,
- and downstream reliance occurring without awareness of material constraints.

Validation Limitation Principle

A limitation does not automatically mean that validation is invalid.

A limitation defines a condition under which the **meaning, strength, applicability, completeness, or permitted reliance of validation may be constrained.**

VALIDOS® therefore distinguishes:

Validation Limitation ≠ Validation Failure

but also:

Undisclosed Limitation ≠ Trustworthy Validation

VLGM requires material limitations to remain part of the governed validation record.

Limitation Categories

Validation limitations may arise from multiple sources.

Scope Limitations

Material dimensions of the Validation Object are intentionally or necessarily excluded from examination.

Access Limitations

Required systems, records, environments, interfaces, personnel, data, or other validation-relevant resources cannot be fully accessed.

Evidence Limitations

Available evidence may be incomplete, insufficiently representative, temporally restricted, or otherwise constrained.

Source Limitations

Sources may contain uncertainty, incomplete provenance, limited independence, restricted accessibility, or other conditions affecting their usefulness.

Methodological Limitations

The selected validation method may have known boundaries, assumptions, sensitivities, or conditions limiting what it can establish.

Environmental Limitations

Validation may be conducted only within a restricted set of technical, physical, organizational, operational, or social environments.

Population Limitations

Validation may cover only defined populations, user groups, subjects, datasets, or affected entities.

Temporal Limitations

Validation may represent only a defined period, historical state, operating window, or lifecycle phase.

Dependency Limitations

The Validation Object may rely upon external systems, infrastructure, services, datasets, actors, or conditions that cannot be completely validated within the current process.

Uncertainty Limitations

Residual uncertainty may remain despite otherwise sufficient validation activity.

Limitation Materiality

Not every limitation requires the same governance response.

VLGM evaluates whether a limitation is materially capable of affecting:

- Validation Purpose,
- Validation Scope,
- required Validation Depth,
- validation criteria,
- validation methodology,
- evidence fitness,
- source reliability,
- validation execution,
- validation determination,
- Validation State,
- Validation Applicability,
- or permitted reliance.

Material limitations must remain visible to all subsequent VALIDOS® governance stages they affect.

Minimum Implementation Framework

1. Identify Validation Limitations

Determine which known conditions constrain validation or the interpretation of its eventual result.

2. Classify Limitations

Classify each material limitation according to its origin and governance relevance.

3. Assess Limitation Impact

Determine whether the limitation affects:

- validation completeness,
- validation confidence,
- applicability,
- interpretation,
- determination,
- state,
- or reliance.

4. Establish Governance Response

For each material limitation, determine whether validation should:

- proceed without modification,
- proceed with documented conditions,
- require additional validation activity,
- require scope modification,
- require depth escalation,
- restrict applicability,
- restrict reliance,
- remain pending,
- or stop until the limitation is sufficiently addressed.

5. Preserve Limitation Traceability

Maintain:

- limitation description,
- limitation category,
- materiality determination,
- affected validation dimensions,
- governance response,
- responsible reference where applicable,
- limitation changes,
- resolution status,
- timestamps,
- and sufficient traceability throughout the validation lifecycle.

Limitation Disclosure

Material limitations must remain available wherever their omission could cause a validation result to be materially misunderstood.

Disclosure should be proportionate to the significance of the limitation and the context in which validation is interpreted.

A validation determination should not be presented in a manner that creates an impression of certainty, completeness, applicability, or coverage that the governed validation does not support.

Limitation Propagation

A limitation identified at one stage of VALIDOS® may affect later stages.

For example:

Access Limitation → Evidence Constraint → Reduced Validation Coverage → Conditional Determination → Restricted Applicability → Restricted Reliance

VLGM therefore requires material limitations to propagate through the validation lifecycle where their effect remains relevant.

A limitation must not disappear merely because validation has progressed to another Parent Standard.

Limitation Resolution

Where a limitation is removed or materially reduced, its governance status should be updated.

Resolution may occur through:

- additional access,
- new evidence,
- improved source quality,
- additional testing,
- methodological supplementation,
- expanded population coverage,
- extended observation,
- dependency validation,
- or another governed corrective action.

Resolution does not automatically modify previous validation determinations.

Where the limitation materially affected earlier validation activity, additional validation or revalidation may still be required.

Residual Limitations

Some limitations may remain after validation is completed.

These **Residual Validation Limitations** should remain attached to the resulting validation record, determination, state, and relevant reliance conditions.

Residual limitations allow downstream users to distinguish between:

- what validation established,
- what validation did not establish,
- and what remains uncertain.

Limitation Change Governance

Limitations may appear, disappear, increase, or decrease throughout the validation lifecycle.

Material change may require reassessment of:

- Validation Scope,
- Validation Depth,
- applicability,
- criteria,
- methodology,
- evidence,
- execution,
- determination,
- state,
- or reliance.

VLGM therefore maintains limitation governance as a continuing responsibility rather than a one-time disclosure activity.

Use Case 1 — AI System with Restricted Training Data Access

Scenario

An external organization validates an AI system but cannot access portions of the proprietary training dataset.

Application

VLGM records the access limitation, determines which validation questions are affected, and establishes whether validation may proceed under documented conditions.

Result

The resulting validation cannot be represented as if unrestricted access to the training data existed. The limitation remains visible wherever it materially affects interpretation or reliance.

Use Case 2 — Predictive Model with Limited Population Coverage

Scenario

A predictive model is evaluated using evidence representing only part of the population for which future deployment is being considered.

Application

VLGM identifies the population limitation and connects it to Validation Applicability.

Result

Validation may remain legitimate for the examined population while broader deployment is prevented from inheriting unsupported validation confidence.

Canonical Closing Statement

Validation Limitation Governance Module establishes the canonical governance framework for preserving the boundaries of what validation can legitimately establish. By governing limitation identification, classification, materiality, impact, disclosure, propagation, resolution, residual uncertainty, change, and traceability, VLGM ensures that validation confidence never silently exceeds the conditions, evidence, access, coverage, and methodological support upon which that confidence depends.

Parent Resources

→ VALIDOS® Validation Governance Architecture — Validation Governance Layer

→ VALIDOS® Architecture Map

→ VALIDOS® Complete Standards & Modules Index

Internal Module Flow

→ Previous module: Validation Applicability Mapping Module (VAMM)

Related Documents

→ Validation Purpose & Scope Standard

→ About Validation Purpose & Scope Standard

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Canonical Source

https://originopenfoundation.org/